INVERTIS UNIVERSITY BAREILLY

Established by Govt. of U.P. u/s 2F of UGC Act, 1956 vide U.P. Act 22 of 2010.

BCS-052	FUZZY LOGIC	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives: To provide a thorough understanding of the internals of Compiler Design

CO1	Develop the skills to gain a basic understanding of neural network theory and fuzzy logic theory
CO2	Introduce students to artificial neural networks and fuzzy theory from an engineering perspective
CO3	Provide an understanding of the basic mathematical elements of the theory of fuzzy sets.
CO4	Explain the concepts of neural networks, fuzzy logic, and genetic algorithms
CO5	Provide an emphasis on the differences and similarities between fuzzy sets and classical sets theories.

MODULE-I

Introduction to fuzzy set theory: Probabilistic reasoning, Fuzzy sets, mathematics of fuzzy set theory, operations on fuzzy sets, comparison of fuzzy and crisp set theory.
Fuzzy mapping: One to one mapping, max-min principle, extension principle, implication rules – mamdani implications.
Membership functions: Universe of discourse, mapping inside fuzzy domain, fuzzy membership mapping methods, and application to real world problems.

MODULE-II

Fuzzy knowledge based systems: Fuzzification, Fuzzy knowledge base, rule base, Data base for fuzzy, Inference rules, defuzzification methods of defuzzification.
Fuzzy controller, Control strategies, general PID controller, Implementation of fuzzy systems in control ,Direct fuzzy controller, Fuzzy P, PI and PID controller, Indirect fuzzy controller – fuzzy in handling the inner loops of control systems.

MODULE-III

Nonlinear systems and adaptive fuzzy controller: Nonlinear systems, modification in fuzzy systems for nonlinear control, Adaptive control, Adaptive control using fuzzy, fuzzy sliding mode controls.
Hybrid systems: Neuro- fuzzy and fuzzy genetic systems, applications to engineering problems.

Text Books:

1. Fuzzy control –Drianlcov.
2. Fuzzy modeling and control –Yager.

Reference Books:

1. Timothy J. Ross, "Fuzzy Logic with Engineering Applications" Wiley India.

CO1	Students will be able to interpret fuzzy set theory and uncertainty concepts
CO2	Students will be able to identify the similarities and differences between probability theory and fuzzy set theory and their application conditions
CO3	Students will get the concept of fuzzy relations
CO4	Students are introduced to fuzzy logic
CO5	Students will be able to apply fuzzy set theory in modeling and analyzing uncertainty in a decision problem.